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Prof. Yaser Jararweh
Editor-in-Chief
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Dear Prof. Jararweh,

Please find enclosed our manuscript, “**Self-Referential Anytime-Valid Byzantine-Robust Aggregation for Federated Intrusion Detection**,” which we submit for consideration for publication in *Cluster Computing*.

Federated learning is a high-performance distributed computing paradigm in its own right: a central aggregator coordinates many autonomous client nodes across communication rounds to train a shared model without pooling raw data, exactly the class of network-of-computers problem this journal covers. That coordination model creates a distributed-systems dependability problem, though, since the aggregator has no inherent way to tell a legitimate client update from one crafted by a compromised or malicious node, and this is especially consequential for federated network-intrusion detection, where an attacker has a direct incentive to degrade the very system meant to catch it. Byzantine-robust aggregation is the standard defense, and a recent line of work brings anytime-valid sequential testing to this problem, offering a formal, cumulative statistical guarantee on false-exclusion behavior that ordinary per-round heuristics cannot provide. In developing such a defense, we identified and formally characterized a previously undiagnosed failure mode: an anytime-valid test that compares a client only against the population is mathematically guaranteed to eventually exclude any client whose behavior sits persistently off-center, so it cannot distinguish an honest node in a heterogeneous cluster from a persistent attacker – and the failure grows worse, not better, as the cluster becomes more homogeneous, the opposite of the intuitive expectation.

Our manuscript introduces ARMOR-FL, which corrects this failure with a self-referential shift test, a scale-floored population statistic, and a threshold-calibrated trust weight, and validates the fix on isolated synthetic reproductions and on three real intrusion-detection benchmarks (CICIDS2017, CICIDS2018, and CICIOT2023) under three attack types of increasing subtlety. We report our results plainly: ARMOR-FL does not outperform the simpler Krum baseline in aggregate, and we discuss at length why, and under which specific cluster conditions the reverse holds. We believe this combination – a formally grounded robustness mechanism for federated distributed computing, an honest diagnostic account of where it does and does not help, and a concrete deployment guideline – makes a genuine contribution to the growing body of federated-learning research this journal has published, including recent work on Byzantine-robust and privacy-preserving federated frameworks (RPCFL) and lightweight federated intrusion-detection backbones (FedSE-1DSqueezeNet), both of which we build on or differentiate against directly.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and this submission. We have no conflicts of interest to disclose. The full implementation, experiment configuration grids, and reassembly-only dataset archives are released alongside the manuscript to support independent verification and extension.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Quang-Vinh Dang
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